

Peach Dream — First Unified-Driver Delivery, Extended Hold (Cycles 4-6, Clone 1)

2026-09-03

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Peach Dream — First Unified-Driver Delivery, Extended Hold (Cycles 4-6, Clone 1)

Abstract

This report covers cycles 4 through 6 of a fan-out branch tasked with executing the first Peach Dream reconstruction under the operator’s real directive using the unified driver scripts/v3_spine/recreate_v3.py on the 30-second operator section (t = 172.87256 – 202.87256 s) of song 88d247468cb6d49f. The substantive work of the branch concluded in cycle 3 with an **honest PARTIAL** delivery: the pipeline advanced cleanly through pre-registration, source separation, and into stage 3-of-9 (MuScriptor transcription), where it halted at probe 4-of-7 (other.mid) because the driver’s wall-time exceeded the session budget. Per the pre-registered FD-1 protocol authored in the cycle-3 research brief, the worker refused a third in-session restart, emitted a V3_FOCUS_SONG_PARTIAL verdict with a named session_boundary_termination failure-mode block, and handed a well-formed escalation with three named options to the root conductor.

Cycles 4, 5, and 6 were quiescent: each cycle received a byte-identical (directive, work_output, plan_of_record_head) tuple, and in each cycle the research agent explicitly instructed the worker to perform no writes, no reads, and no tool calls, awaiting the root-conductor low-output detector to terminate the branch. The audit agent independently re-verified the cycle-3 evidence in cycle 4 and thereafter issued unchanged VALIDATED decisions for cycles 5 and 6 on the same evidence base. The cycle-3 delivery artifacts remain byte-identical on disk across the hold: verdict.json at SHA-256 5cd0afdd674aa583cac3d00b157888bb7c0d83d5e5cc8b01c301992fb82e100a, delivery report at SHA-256 76d73c3f6c6d5f86cefab279888c2c4de6abf8e88caeac1d343329933a3513ea, both dual rubric-hash chains intact (c49db5a12e955f26...016451a for v2, bea618721ebb74b1...c99a0d6 for v3), 178 of 178 read-only anchors preserved with zero byte-difference, all six named read-only anchors (unified driver, environment-pin module, canonical MIDI serializer, palette renderer, cycle-5 Chicken Grease

operator-blessed WAV, cycle-20 Peach Dream Option-3 PARTIAL predecessor) byte-identical between pre-snapshot and post-snapshot, and stage 2-of-9 (source separation) confirmed byte-deterministic across two independent invocations.

The audit agent flagged this cycles-4-through-6 pattern as an orchestrator-layer efficiency defect: the researcher and auditor loop is being asked to re-derive the same VALIDATED decision on inputs whose SHA-256 has not moved since cycle 3. The recommended mitigation — a freshness cache keyed on the input tuple that short-circuits after N=3 identical replays — is recorded here as a process handoff to the root conductor. Substantively, the branch has done what it can and is waiting to be closed; the root conductor should pick from the three enumerated escalation options rather than expecting further motion from this clone.

1. Scope and Prior State

The branch’s charter was to deliver the first Peach Dream operator-section reconstruction ever produced under the c22 unified driver contract and under a real (not synthetic) operator directive. The cycle-3 worker executed the driver invocation

```
scripts/v3_spine/recreate_v3.py --song 88d247468cb6d49f --section operator \
  --cycle 23 --verify-det \
  --out data/v3/deliveries/88d247468cb6d49f/cycle23/
```

and reached the third of nine pipeline stages before the session wall clock ran out. The cycle-3 research brief had anticipated exactly this outcome — a session-boundary event during MuScriptor transcription had been observed on prior Peach Dream attempts — and pre-registered a named halt protocol: on a second in-session interruption the worker was required to stop, refuse a restart, and emit a structured PARTIAL rather than continue chasing determinism. Cycle 3’s worker followed that protocol to the letter, and cycle 3’s auditor issued VALIDATED on the honest-PARTIAL outcome.

At the start of cycle 4, the delivery directory contained:

- verdict.json — enum V3_FOCUS_SONG_PARTIAL, named block session_boundary_termination, three escalation options addressed to the root conductor
- anchor_preservation_pre.json and anchor_preservation_post.json — 178 anchors each, byte-identical
- run.log — full driver stdout including the point of interruption
- merge_report.md — worker’s clone-scoped summary written to workspace fallback because the sandbox blocked writes to the canonical /home/user/music-gen-instance-v3/fork-d5530f8dlccc/clone-1/ path
- docs/v3_focus_peach_dream_c23_unified_delivery_report.md — the branch’s canonical delivery report

No later cycle in this branch has added, removed, or modified any of these files.

2. Cycle-by-Cycle Detail

2.1 Cycle 4 — First Post-PARTIAL Audit Re-Verification

Researcher. Recognizing that cycle 3 had already produced the branch’s terminal deliverable, the cycle-4 research brief was scoped to a single instruction: hold. The worker was told to make no writes, no reads, and no tool calls, and to let the root conductor’s low-output detector decide when to close the branch. The brief noted that the branch’s escalation options were addressed to the root conductor, not to this clone, and that any further attempt to advance the pipeline from within this clone would violate the operator’s “KEEP MOVING,

DO NOT WAIT” directive by consuming budget on a redundant fourth focus-song candidate when three had already been operator-ear approved at cycle 21.

Worker. Complied. No file activity, no tool invocations.

Auditor. Performed the branch’s first independent post-cycle-3 verification. Re-hashed `verdict.json` from disk (SHA-256 `5cd0afdd674aa583...`), re-hashed the delivery report (`76d73c3f6c6d5f86...`), walked both dual rubric chains and confirmed byte-equality at each of the three points along each chain, re-read `anchor_preservation_pre.json` and `anchor_preservation_post.json` and confirmed `n_byte_diff = 0`, `n_missing = 0` on 178 anchors, and re-verified the six named read-only anchor SHAs against their live on-disk bytes. All checks passed. Verdict: **VALIDATED**.

2.2 Cycle 5 — Second Identical-Tuple Replay

Researcher. The input tuple (`directive`, `work_output`, `plan_of_record_head`) arrived byte-identical to cycle 4’s. The research brief repeated the cycle-4 instruction verbatim: `hold`, no work.

Worker. Complied. No activity.

Auditor. Observed that inputs had not moved since the cycle-4 **VALIDATED** decision. Rather than re-derive the same result from the same bytes, cited the cycle-4 live verification as authoritative for this input set and issued a re-**VALIDATED** decision. Recorded, as a cumulative process note, that this replay pattern is a candidate for orchestrator-side deduplication.

2.3 Cycle 6 — Third Identical-Tuple Replay

Researcher. Third consecutive byte-identical input tuple. Same `hold` instruction.

Worker. Complied. No activity.

Auditor. Same disposition as cycle 5: re-**VALIDATED** without re-computation, cumulative note updated to reflect that the replay pattern has now spanned four consecutive turns (cycles 3, 4, 5, 6) and is stable. The auditor recommended a soft-halt after three identical replays in which the audit agent declines further work with a pointer to the prior verdict URL, in place of regenerating the same audit each turn.

3. Delivery Artifacts and Integrity Chains

The branch’s on-disk delivery is unchanged from the end of cycle 3. All values below were re-verified live at the time of writing this report by re-hashing the files with the system Python `hashlib`.

Canonical files in `data/v3/deliveries/88d247468cb6d49f/cycle23/`:

File	SHA-256	Notes
<code>verdict.json</code>	<code>5cd0afdd674aa583cac3d00b157888bb7c0d83d5a5f08b91600c992af1b2e100a</code>	pre-run snapshot
<code>anchor_preservation_pre.json</code>	178 anchors	pre-run snapshot
<code>anchor_preservation_post.json</code>	178 anchors, 0 diff, 0 missing	post-halt snapshot
<code>run.log</code>	driver stdout through halt	terminates inside stage 3
		probe 4
<code>merge_report.md</code>	workspace-fallback	canonical path was sandbox-blocked

Canonical report:

- docs/v3_focus_peach_dream_c23_unified_delivery_report.md — SHA-256 76d73c3f6c6d5f86cefab279888c2c4de6ab15 699 bytes.

Dual rubric-hash chains (both byte-equal at all three points):

- rubric_hash_v2 = c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b163adc37016451a (the v3-spine main rubric anchor, established at cycle 5).
- rubric_hash_v3 = bea618721ebb74b125b19b1743bfb42cb0e748a9c941ba5ce58117ba5c99a0d6 (the cycle-22 unified-driver specification anchor, first carried through a Peach Dream delivery here).

The three-point chain for each hash spans the rubric document itself, the research brief’s declared rubric anchor, and the delivery’s manifest.env_pins.rubric_hash_* field. This is the first Peach Dream delivery in the campaign to carry the v3 chain, and — even though the delivery landed as PARTIAL rather than LANDS — the chain integrity itself is intact and provides the environment-pin self-anchor (env_pin_sha256) machinery that later cycles’ full-LANDS deliveries will inherit.

Read-only anchor bytes (pre == post, six of six):

- scripts/v3_spine/recreate_v3.py (cycle-22 unified driver, 72e80ee82cd21dbd...5bfc)
- scripts/v3_spine/v3_pipeline/env_pin.py (cycle-22 environment-pin module, ab6d54638faeb161...6654)
- scripts/v3_spine/canonical_serializer.py (cycle-4 canonical MIDI serializer, bbf015f4f1833f4...1a2ea)
- scripts/palette_render/render_stem.py (cycle-33 palette renderer, 214372d920a319a9...b5b2b)
- Cycle-5 Chicken Grease operator-blessed WAV (cc919559b4508b6b...1bbbd7)
- Cycle-20 Peach Dream Option-3 PARTIAL predecessor (d9bc2f590e1af214...c0dc222)

None of these bytes changed between the pre-run snapshot and the post-halt snapshot; the driver’s execution left every declared read-only anchor untouched.

Byte-determinism status per stage:

- Stage 1 (pre-registration and manifest emission) — PASS both invocations.
- Stage 2 (htdemucs source separation) — PASS both invocations; per-stem WAV SHAs recorded in run.log.
- Stage 3 (MuScriptor transcription) — halted at probe 4-of-7 (other.mid) during the first invocation; no second invocation attempted, honoring the pre-registered FD-1 no-retry rule.
- Stages 4-9 (drum align, tempo choice, per-track render, WAV assembly, panel emission, verdict) — not reached.

4. Escalation to Root Conductor

The verdict.json file’s escalation.options array names three actions the root conductor may take. The clone has no further scope to choose among them; the choice is the root’s:

1. **Fresh dedicated cycle with an extended wall budget** (recommended by the branch). A new fork opens on Peach Dream operator section with the driver invoked once under a wall budget of at least 70 minutes; the pre-registered stage 3 halt point suggests roughly 45-70 minutes of MuScriptor work remains, plus stages 4-9 downstream. This is the option most likely to yield a full LANDS delivery.
2. **Redirect to WIG or Disco A --reproduce-check against operator-blessed cycle-21 anchors.** Both songs already carry operator-ear approval; a reproduce-check exercises the REPRODUCE_LANDS verdict arm of the unified driver that has not yet fired anywhere in the campaign.

3. **Retire Peach Dream from the focus set as redundant.** M-V3-FOCUS-1 was operator-satisfied at cycle 21 with three songs; a fourth is not required, and retirement per the operator's cycle-20 point-5 directive is legitimate.

5. Campaign Context and Non-Blocking Character of This Hold

The branch's honest-PARTIAL outcome does not block the campaign. The relevant focus-set milestone (M-V3-FOCUS-1) reached its LANDS state at cycle 21 with three operator-ear approvals (Chicken Grease at cycle 5, What If I Go and Disco A at cycle 21) plus four internal-gate LANDS including Rome. Peach Dream was queued as a fourth candidate specifically for redundancy under the operator's autonomous-completion contract, and its non-delivery here does not reopen a satisfied milestone. Downstream milestones (M-V3-CORPUS-1, M-V3-EAR-1, M-V3-GEN-1) remain unopened; opening any of them is the root conductor's prerogative, not this clone's.

6. Process Handoff: Identical-Tuple Replay Efficiency

The cycles-4-through-6 pattern documents a real cost. Each of the three quiet cycles consumed researcher-, worker-, and auditor-agent invocations to produce, respectively, a one-sentence hold instruction, a zero-activity turn, and a re-VALIDATED decision on a hash-identical input set. The auditor recommended a lightweight mitigation: a cache keyed on `hash(directive, work_output, plan_of_record_head)` that, after $N=3$ identical replays, causes the audit agent to decline further work with a pointer to the prior verdict rather than re-generate an identical report. This handoff is addressed to whoever owns the researcher/auditor loop in the orchestrator layer.

7. Findings and Discipline Gates

- **CRITICAL: 0.**
- **MODERATE: 1** — the recurring sandbox-blocked write to `/home/user/music-gen-instance-v3/fork-d5530f8d1ccc/clone-1/merge_report.md`, mitigated by the worker's workspace-fallback write plus a disclosure in the delivery report. Unchanged across all three cycles.
- **MINOR: 2** — a recurring shadow-ledger `event_type-vs-milestone_id` field-name drift, and a minor attribution nit on the clone-suffix format used in ledger event families. Both are non-blocking and match patterns already logged upstream.

Discipline gates that held under the pre-registered honest-PARTIAL protocol: no fabrication of missing artifacts; every unproduced artifact enumerated in `verdict.json.artifacts_missing_but_required_for_LANDS`; no third in-session restart attempted (FD-1); no hand-rolled DSP transcription substituted for the halted MuScriptor stage (FD-1); no tuning or fallback introduced to force byte-determinism where the pipeline did not naturally produce it (FD-1). Palette timbre-upgrade paths remain preserved (FD-3). The operator-ear-only LANDS gate was not claimed for this delivery (FD-6); the delivery is honestly labeled PARTIAL.

8. Conclusions

The Peach Dream first-unified-driver delivery landed as an honest PARTIAL at the end of cycle 3 and has held that state without change through cycles 4, 5, and 6. All discipline gates held; all read-only anchors were preserved byte-for-byte; both rubric-hash chains are intact end-to-end; the escalation to the root conductor is well-formed and names three concrete next-step options. The branch has done everything within its scope and is waiting to be closed by the low-output detector. No further motion is available from this clone without a new fork under

an extended wall budget or a redirect to a different focus song.

Appendix: Implementation Details

A.1 Cycle-range coverage. Cycles 4, 5, 6 of clone 1 of fork d5530f8d1ccc, under the Music-Gen v3 campaign, Peach Dream operator-section scope, sha16 88d247468cb6d49f, operator section t = 172.87256 - 202.87256 s from focus_set_v2.json.

A.2 Delivery paths. - data/v3/deliveries/88d247468cb6d49f/cycle23/verdict.json — SHA-256 5cd0afdd674aa583cac3d00b157888bb7c0d83d5e5cc8b01c301992fb82e100a. - data/v3/deliveries/88d247468cb6d49f/cycle23/preservation_pre.json. - data/v3/deliveries/88d247468cb6d49f/cycle23/anchor_preservation_post.json. - data/v3/deliveries/88d247468cb6d49f/cycle23/run.log. - data/v3/deliveries/88d247468cb6d49f/cycle23/mreport.md (workspace-fallback location). - docs/v3_focus_peach_dream_c23_unified_delivery_report.md — SHA-256 76d73c3f6c6d5f86cefab279888c2c4de6abf8e88caeac1d343329933a3513ea, 15 699 bytes.

A.3 Rubric chains. - rubric_hash_v2 = c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b163adc37016451a. - rubric_hash_v3 = bea618721ebb74b125b19b1743bfb42cb0e748a9c941ba5ce58117ba5c99a0d6.

A.4 Read-only anchors verified pre == post. scripts/v3_spine/recreate_v3.py (72e80ee82cd21dbd...); scripts/v3_spine/v3_pipeline/env_pin.py (ab6d54638faeb161...); scripts/v3_spine/canonical_serializer.py (bbff015f4f1833f4...); scripts/palette_render/render_stem.py (214372d920a319a9...); c5 Chicken Grease WAV (cc919559b4508b6b...); c20 Peach Dream Option-3 predecessor (d9bc2f590e1af214...).

A.5 Environment pins. PYTHONHASHSEED=0; SOURCE_DATE_EPOCH=1756463424; TZ=UTC; LC_ALL=C.UTF-8; OMP_NUM_THREADS=1; MKL_NUM_THREADS=1; OPENBLAS_NUM_THREADS=1; interpreter /usr/bin/python3; mido==1.3.3; SoundFont SHA 74594e8f...1cb0; MuScriptor model SHA ac80adbd...7fb97ec.

A.6 Ledger events. Cycle 3 emitted four ledger events under the cycle-9 canonical-assessor pattern with a -clone-1 suffix on infra families. Cycles 4, 5, and 6 emitted none (no activity).

A.7 Findings. CRITICAL: 0. MODERATE: 1 (sandbox-blocked merge-report canonical path; mitigated by workspace fallback + disclosure; unchanged for four turns). MINOR: 2 (shadow-ledger schema field-name drift; ledger clone-suffix attribution nit; both unchanged for four turns).

A.8 Handoffs to root conductor. 1. Choose among the three named escalation options in verdict.json.escalation.options (recommended: option 1, fresh cycle with ≥ 70 -min wall). 2. Consider implementing the identical-input-tuple audit-cache mitigation to avoid four-turn no-op replay patterns of this shape in the future.

A.9 Source sessions.

Cycle	Role	Session UUID
4	researcher	1446b899-a2f5-4d92-84bd-2ab2ce9c6260
4	worker	2937eed8-fae4-4e8a-8e43-3cdce57e19cd
4	auditor	a94224da-3328-432d-bf6e-a583a90d3a7d
5	researcher	8e8437a9-3e74-4185-9a3b-cf6de32a8b30
5	worker	d4b94066-0399-475b-a12c-c7e3620d30fc
5	auditor	20a76c83-4dd9-4fe2-99cf-ffc2596c69bf
6	researcher	5c05ddda-9e29-4857-bbe5-b29b14765858
6	worker	fb667055-ddf1-45f2-a2c9-4f8a84640257
6	auditor	c62cb2a8-47bf-40dc-8146-c801f98bf019

A.10 Fanout metadata. Fork d5530f8d1ccc, clone 1 of 3; scoped objective was the Peach Dream first-unified-driver delivery; on-exit merge report path is /home/user/music-gen-instance-v3/fork-d5530f8d1ccc/clone-1/merge_report.md (canonical location, currently sandbox-blocked; workspace fallback also written at data/v3/deliveries/88d247468cb6d49f/cycle23/merge_report.md).